Name: Leah Weldon
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Role: Project Code Owner

For this group project, my role was to ensure that the overall application functions as expected by testing the various versions of the code and I also contributed to the code itself by writing functions and helping to implement the basic specifications.

1) Project Code Owner Responsibility and Testing:

   For this project as the code owner I was in charge of conducting the testing of the application to ensure it was running as expected. We had three different versions of our code and for weeks 2, 3 and 4 I would test each. I was looking for display errors, number inconsistencies, unexpected loops and incorrect outputs. I would test the code with different possible player input and ensure that the correct output was returned. Where possible I rectified these errors with the assistance of other members who may have written the various functions. For week 4 I tested the code after every new commit to ensure no unexpected errors would arise so close to the deadline.

2) Validating Answers:

   This function was written to be used with the *display_input()* function. I created a function called *validate_input_sequence()* to compare the user's input to the correct answer. I use the *strcmp()* function to compare the stored user input to the Morse code of the letter currently being tested in the ‘alphabet’ array. If both strings are identical then I return 1 and otherwise I return 0 to indicate the answer is incorrect.

3) Setting and updating the LED Colour:

   I created and implemented functions to initialise and then change the colour of the LED within the Morse code game. The first function *start_RGB_Colour()* sets the initial colour of our LED to be blue and this indicates the game has yet to begin. The next function *RGB_update()* updates our LED depending on the number of lives the player has remaining. I used a switch statement to update the colour with either black (off) for 0 lives, red for 1 life remaining, yellow for 2 lives and green for all 3 lives remaining. I used the *put_pixel()* function to set the LED. I also dealt with some error handling within this function.

4) Validating User Input:

   I also wrote another function that was used for validating input and dealing with input handling. For this function I used a loop to iterate over the set of alphanumeric characters, comparing them to the user input sequence. I check if the user input matches our Morse code for our current character. If a match is found then I return the index of the matched character in the alphabet array and if not, then I return -1.

5) Code Documentation
   I took on the responsibility to write the high level explanation of our code in the group report documentation. I broke down our workload and explained in detail each aspect of our working code and when it was implemented over the 4 week duration. I explained each of the main functions we created and their purpose.

In Conclusion, we have successfully created our Morse Code game that meets all of the requirements and objectives that we set out to achieve, including the optional extra features. We effectively collaborated together and communicated efficiently to create this fun and engaging game. Overall, this project has been a great learning experience and it is very rewarding to see the final product after all the time we dedicated to it. It has greatly improved my teamwork and github use and I am very proud of the work we were able to produce.